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Complete Reinsurance Contract Type Inventory & Simple Calculation Reference

A pure reference to reinsurance/ILS structures and how each is calculated — no system implementation details. Sourced primarily from AIR Worldwide’s *Quick Reference for Analyzing Reinsurance Contracts* (authoritative for the mechanics marked “confirmed”), supplemented with standard market practice for structures that document doesn’t cover (marked “market standard”).

Part 0: Foundational Mechanics (apply across most structures)

Occurrence vs. Aggregate Terms — the order of application (confirmed)

Every layered structure applies terms in this exact order:

Step	Term	What it does
1	Occurrence Retention	If the event’s loss $\leq$ retention, everything is retained — stop here.
2	Occurrence Limit	If loss exceeds retention, the eligible loss for that event is capped at the limit. Anything above the limit is retained.
3	Aggregate Retention	The eligible loss (from step 2) is reduced by whatever aggregate retention remains, eroding it event by event until exhausted.
4	Aggregate Limit	Whatever survives step 3 is paid, up to the annual aggregate limit.

Simple formula:

Eligible Loss (per event) = $\min(\max(0, \text{Event Loss} - \text{Occ Retention}), \text{Occ Limit})$

Aggregate Retention Remaining = $\max(0, \text{Aggregate Retention}$

$-\sum \text{prior Eligible Losses this year})$

Contract Loss (per event) = $\min(\max(0, \text{Eligible Loss} - \text{Aggregate Retention Remaining}), \text{Aggregate Limit Remaining})$

Reinstatement & Reinstatement Premium

A **reinstatement** restores a used-up occurrence limit so the layer can respond to a later event in the same period. “0 reinstatements” = the layer pays once, period. “1 reinstatement” = it can respond fully twice. “Unlimited” (sometimes coded 99) = it can respond every time, capped only by the number of events.

Simple formula:

Aggregate Limit = Occurrence Limit × (1 + Number of Reinstatements)

Reinstatement Premium (market standard — not explicitly formulated in the AIR doc, but this is the standard actuarial approach) is the additional premium charged for restoring the limit, usually pro-rated by (a) how much of the limit was used and (b) how much time remains in the period:

Reinstatement Premium = Original Premium
× (Limit Reinstated / Full Occ Limit) [pro rata as to amount]
× (Time Remaining / Full Period) [pro rata as to time]
× Reinstatement Provision % [100%, 75%, 50%, "free"]

A reinstatement provision of “100%” means the reinstatement costs as much as the original premium (scaled by amount/time); “free” reinstatements cost nothing.

Coinsurance, Participation, and Percent Placed (confirmed)

These three terms sound similar but do different things:

Term	What it does	Formula
Insurer's Coinsurance	The cedant retains this % of the <i>layer</i> loss itself	Reinsurer's share = Layer Loss × (1 – Coinsurance %)
Percent Placed	The % of the treaty actually placed with reinsurers — affects only <i>net retained loss</i> , not the 100%/Gross/Net contract loss figures	Net Retained Loss = Company Loss – Inuring Program Loss – (100% Contract Loss × % Placed)
Gross / Net / Participation 3 / Participation 4	Scales the 100% contract loss down to one reinsurer's specific share (e.g., for retrocession)	Reinsurer's Loss = 100% Contract Loss × Participation %

Order matters: Coinsurance is applied to determine the layer's total reinsured loss; participation percentages then scale that down to an individual reinsurer's share; percent placed only affects what's left as the cedant's *net retained* loss, not the contract loss itself.

Inuring vs. Sourcing Relationships (confirmed)

These describe how multiple programs interact:

- **Sourcing:** Program A's *loss* becomes the input to Program B (e.g., a “company retention” program calculates net retained loss, which then sources into a catastrophe program as its starting loss).
- **Inuring:** Program A responds *first*, and its recovery is subtracted from the loss before Program B (which “is inured to the benefit of” A, or equivalently A “inures to” B) ever sees it. This is the mechanism behind deductibles-before-a-layer, cascading layers, and top-and-drop structures.

Part 1: Proportional (Pro Rata) Structures

1.1 Quota Share (QS) — confirmed

The reinsurer takes a fixed percentage of every loss (and premium) in the covered book, with an optional occurrence and/or aggregate cap.

Simple formula:

Contract Loss = $\min(\text{Ceded \%} \times \text{Company Loss}, \text{Occurrence Limit})$ [per event]
Contract Loss (year) = $\min(\sum \text{above}, \text{Aggregate Limit})$ [annual cap]

Ceded % is set per geography/area if needed (e.g., 20% of US business only).

1.2 Surplus Share — market standard

Like quota share, but the cession % *varies per risk* based on a retained “line” — the cedant keeps a fixed dollar amount per risk and cedes the rest, up to a multiple of that line.

Simple formula:

Cession % (per risk) = $\max(0, (\text{Risk Sum Insured} - \text{Retained Line}) / \text{Risk Sum Insured})$
capped at: $(\text{Number of Lines} \times \text{Retained Line}) / \text{Risk Sum Insured}$
Reinsurer's Loss = Cession % $\times$ Risk-Level Loss

Unlike quota share, two risks of different sizes in the same treaty get different cession percentages.

Part 2: Non-Proportional (Excess of Loss) Structures

2.1 Per-Risk Excess of Loss — market standard

Protects against a large loss on any *single risk*, independent of catastrophe.

Simple formula:

Recovery = $\min(\max(0, \text{Single-Risk Loss} - \text{Retention}), \text{Limit})$

2.2 Per-Occurrence (Catastrophe) Excess of Loss — confirmed

The core CAT reinsurance structure — protects the whole book against one occurrence.

Simple formula:

Recovery = $\min(\max(0, \text{Occurrence Loss} - \text{Occ Retention}), \text{Occ Limit})$

Layered towers stack multiple XOL layers with increasing retentions (e.g., \$50M xs \$50M, then \$75M xs \$100M, then \$100M xs \$175M) — coinsurance can apply independently to each layer.

2.3 Aggregate Excess of Loss — confirmed

Protects against the *cumulative total* of losses in a period exceeding a retention (frequency protection).

Simple formula:

Contract Loss = $\min(\max(0, \sum \text{Occurrence Losses in Period} - \text{Agg Retention}), \text{Agg Limit})$

Note: aggregate terms aren't available on a pure Catastrophe XOL program type — there, the aggregate limit is instead controlled indirectly via the reinstatement count ($\text{Agg Limit} = \text{Occ Limit} \times (1 + \text{Reinstatements})$).

2.4 Excess of Loss with (Outer) Aggregate Deductible – confirmed

The aggregate deductible is applied **before** any per-occurrence terms — it must be eroded first, from the ground up, before the XOL layer even starts responding.

Simple formula (built as two inuring programs):

Step 1 (Inuring "deductible" program, e.g. a Quota Share with an Agg Limit = deductible amount):

Absorbed = $\min(\sum \text{Ground-Up Losses}, \text{Outer Aggregate Deductible})$

Step 2 (the actual XOL layer, receiving the loss net of Step 1):

$$\text{Remaining Loss} = \text{Ground-Up Loss} - (\text{Step 1's absorbed amount for that event})$$

$$\text{Contract Loss} = \min(\max(0, \text{Remaining Loss} - \text{Occ Retention}), \text{Occ Limit})$$

2.5 Excess of Loss with Otherwise Recoverable (Inner Aggregate) Deductible — confirmed

The opposite order from 2.4 — the deductible is applied **after** occurrence terms but **before** the loss is actually paid out. Losses that would otherwise be recoverable under the occurrence terms are absorbed by this inner deductible first.

Simple formula:

$$\text{Eligible Loss (per event)} = \min(\max(0, \text{Occurrence Loss} - \text{Occ Retention}), \text{Occ Limit})$$

$$\text{Aggregate Retention Remaining} = \max(0, \text{Agg Retention} - \sum \text{prior Eligible Losses})$$

$$\text{Contract Loss} = \min(\max(0, \text{Eligible Loss} - \text{Aggregate Retention Remaining}), \text{Agg Limit})$$

This is mechanically identical to the standard 4-step order in Part 0 — “inner aggregate deductible” is just another name for the Aggregate Retention term applied in its normal sequence (as opposed to the “outer” version in 2.4, which sits entirely outside/before the occurrence terms).

2.6 Franchise Deductible (“Disappearing Deductible”) – confirmed

A **threshold**, not a true deductible — once the loss reaches the threshold, the *entire* loss is paid, not just the amount above it.

Simple formula:

```
Contract Loss = 0          if Loss < Trigger Threshold
```

```
Contract Loss = min(Loss, Occ Limit)    if Loss ≥ Trigger Threshold
```

Nothing is subtracted for the threshold itself once it's crossed — that's what makes the deductible "disappear."

2.7 Reverse Franchise Deductible – confirmed

The mirror image of 2.6 — a **ceiling**, not a floor. The contract only pays if the loss stays *below* the threshold.

Simple formula:

$$\text{Contract Loss} = \min(\text{Loss}, \text{Occ Limit}) \quad \text{if } \text{Loss} \leq \text{Trigger Threshold}$$

```
Contract Loss = 0 if Loss > Trigger Threshold
```

2.8 Drop-Down Cover – confirmed

The retention (and often the limit) *lowers* for subsequent events after the first layer is exhausted — modeled as a “step-layer” that activates once the primary layer’s reinstatements run out.

Simple formula:

Layer 1: Recovery = $\min(\max(0, \text{Loss} - \text{Retention}_1), \text{Limit}_1)$ [while Layer 1 has capacity]
 Step-Layer: Recovery = $\min(\max(0, \text{Loss} - \text{Retention}_2), \text{Limit}_2)$
 [Retention_2 < Retention_1, once Layer 1 is exhausted]

2.9 Nth-Event Cover — confirmed (two distinct methods)

Coverage doesn't apply until the **Nth** event in a period — the cedant self-insures the first N–1 events that would otherwise trigger the layer.

Method A — “Otherwise Recoverable” (via Aggregate Retention sized to the occurrence limit):

Set Agg Retention = Occ Limit $\times$ (N – 1)

Contract Loss = same 4-step formula as Part 0 — the aggregate retention absorbs the first N–1 full-limit recoveries before anything is actually paid

Method B — “Starting Event Number” (a hard sequence filter):

Contract Loss = 0 for Event Sequence < N
 Contract Loss = $\min(\max(0, \text{Loss} - \text{Occ Retention}), \text{Occ Limit})$ for Event Sequence $\geq$ N

The two methods differ subtly: Method A only skips events large enough to have used up a full occurrence limit's worth of retention; Method B skips the first N–1 events *entirely*, regardless of size.

2.10 Cascading Structure — confirmed

Multiple layers stacked so that when one layer's limit is exhausted, the excess **feeds into** (sources/inures to) the next — losses already paid by an earlier layer are removed before the next layer's own terms apply.

Simple formula (three cascading layers, illustrative):

Layer 1 Loss = $\min(\max(0, \text{Event Loss} - \text{Ret}_1), \text{Limit}_1)$

Layer 2 Loss = $\min(\max(0, (\text{Event Loss} - \text{Layer 1 Loss}) - \text{Ret}_2), \text{Limit}_2)$

Layer 3 Loss = $\min(\max(0, (\text{Event Loss} - \text{Layer 1 Loss} - \text{Layer 2 Loss}) - \text{Ret}_3), \text{Limit}_3)$

Each layer's own retention is applied fresh to what's left after the prior layer's *paid* loss (not just its retention) is stripped out.

2.11 Top-and-Drop Structure — confirmed

A single top layer that reinstates as a *lower* layer if untouched by the first event — giving the cedant either full protection at the top for one big loss, or double protection lower down across two moderate losses.

Simple formula:

IF First Event Loss exhausts all lower CAT layers:

Top Layer Loss = $\min(\max(0, \text{Remaining Loss} - \text{Top Retention}), \text{Top Limit})$
 [used at the TOP]

IF First Event Loss does NOT exhaust the lower CAT layers:

Top Layer "drops" to act as an additional lower layer for the NEXT event:

Top Layer Loss (2nd event) = $\min(\max(0, \text{2nd Event Loss} - \text{Lower Retention}), \text{Top Limit})$

Modeled either as a step-layer (drops down after the first layer is exhausted) or as a cascading structure (inures to the benefit of the lower layers, engaging only on overflow or on a subsequent event).

Part 3: Index & Parametric Structures

3.1 Industry Loss Warranty (ILW) — confirmed

Pays based on an **industry-wide** loss estimate exceeding a threshold, rather than the cedant's own loss — basis risk (industry vs. own loss) is the defining feature. Modeled as a standard XOL layer, but with an Industry Loss Minimum/Maximum (or Company Loss Minimum/Maximum, if the trigger is measured only within the cedant's own footprint) as the *trigger*, separate from the layer's own retention/limit.

Simple formula:

Trigger Met = (Industry Loss for the event $\geq$ Industry Loss Minimum)

[or: Company Loss (proxy exposure) $\geq$ Company Loss Minimum,
if using a footprint proxy]

Contract Loss = $\min(\max(0, \text{Loss} - \text{Occ Retention}), \text{Occ Limit})$ IF Trigger Met, ELSE 0

Two flavors: **indemnity-based** (pays your own loss once the industry index trips) or **binary** (pays the full limit outright once triggered, regardless of your own loss).

3.2 Parameter-Based ILW — confirmed

Same as 3.1, but the trigger is a **physical event parameter** instead of a loss index — e.g., earthquake moment magnitude ≥ 7.0 with an epicenter in a defined region, or hurricane central pressure/landfall windspeed/Carvill Hurricane Index exceeding a threshold.

Simple formula:

Trigger Met = (Event Parameter meets ALL specified rules,
e.g., Magnitude ≥ 7.0 AND Epicenter in Region X)

Contract Loss = $\min(\max(0, \text{Loss} - \text{Occ Retention}), \text{Occ Limit})$ IF Trigger Met, ELSE 0

Because there's no loss index to reference, exposure is typically represented by a proxy "industry" company built from user-specified or standard market-share (take-up rate) data for the region.

Part 4: Capital Markets / ILS Structures

4.1 CAT Bond — market standard

A fully collateralized instrument: investors post principal; a triggering event reduces principal to pay the sponsor; investors earn a coupon for bearing the risk.

Simple formula:

Coupon = (Reference Rate + Annual Spread %) $\times$ Principal Amount

Principal Reduction on Trigger = $\min(\text{Loss}, \text{Principal Amount})$
[indemnity, industry-index, or parametric basis]

Full Exhaustion = Loss $\geq$ Principal Amount

4.2 Sidecar — market standard

A special-purpose vehicle taking a proportional (quota-share-style) share of a defined segment of business, fully collateralized.

Simple formula:

Investor's Loss = Participation % $\times$ Segment's Net Loss,
capped at the vehicle's Aggregate Limit (= its total collateral)

4.3 Collateralized Reinsurance (CRI) — market standard

Not a distinct attachment structure — it's a *funding* label describing reinsurance written by a fully-collateralized vehicle rather than a rated carrier. Economically, it can be any of the structures above (quota share, per-occurrence XOL, aggregate XOL); only the collateral/trust mechanism differs.

Simple formula: Use whichever formula from Parts 1-2 matches its actual attachment basis (proportional vs. per-occurrence vs. aggregate) — CRI adds no separate calculation of its own.

4.4 RPP — uncertain / not independently confirmed

Seen as a treaty-type code alongside Quota Share and Aggregate XOL in system documentation, but its full expansion and distinct mechanics weren't found in the reference material reviewed. Most likely a proportional or reinstatement-premium-related structure given the company it keeps (grouped with QS and AggXOL) — flagging this honestly rather than guessing at a formula.

Quick-Reference Summary Table

Type	One-line formula
Quota Share	Cession % × Loss, capped by Occ/Agg limits
Surplus Share	Variable Cession % (by risk size) × Risk Loss
Per-Risk XOL	$\min(\max(0, \text{Risk Loss} - \text{Retention}), \text{Limit})$
CAT XOL (per-occurrence)	$\min(\max(0, \text{Occ Loss} - \text{Retention}), \text{Limit})$
Aggregate XOL	$\min(\max(0, \Sigma \text{Loss} - \text{Agg Retention}), \text{Agg Limit})$
Outer Aggregate Deductible	Deductible absorbed <i>before</i> occurrence terms apply
Inner Aggregate (“Otherwise Recoverable”) Deductible	Deductible absorbed <i>after</i> occurrence terms, <i>before</i> payment
Franchise Deductible	0 below threshold; full loss at/above threshold
Reverse Franchise Deductible	full loss at/below threshold; 0 above threshold
Drop-Down Cover	Lower retention/limit kicks in after Layer 1 exhausts
Nth-Event Cover	Skip first N–1 (by retained aggregate or by hard sequence count)
Cascading Structure	Each layer nets out all prior layers' <i>paid</i> loss first
Top-and-Drop	Reinstates as a lower layer if untouched by the first event
ILW	Index/threshold trigger, then standard XOL formula
Parameter-Based ILW	Physical-parameter trigger, then standard XOL formula
CAT Bond	$\min(\text{Loss}, \text{Principal})$; coupon = $(\text{Rate} + \text{Spread}) \times \text{Principal}$
Sidecar	Participation % × Loss, capped at Aggregate Limit (collateral)
Collateralized Re (CRI)	Whichever underlying structure applies — no separate formula
RPP	Not independently confirmed